

$$\Delta(x-y) = i \langle 0 | T \phi(x) \phi(y) | 0 \rangle$$

• Normalize $\phi(x)$ s.t.

$$\begin{cases} \langle 0 | \phi(x) | 0 \rangle = 0 \\ \langle k | \phi(x) | 0 \rangle = e^{-ikx} \end{cases}$$

i.e. $\phi(x)$ is renormalized in the inf. theory

$$\begin{cases} \langle k | k' \rangle = (2\pi)^{d-1} 2\omega \delta^{d-1}(\vec{k} - \vec{k}') \\ \omega = (\vec{k}^2 + m^2)^{1/2} \end{cases}$$

• Completeness relation: $\int \hat{d}k |k\rangle \langle k| = \mathbb{I}$

$$\hat{d}k \equiv \frac{d^d k}{(2\pi)^{d-1} 2\omega}$$

$$\Delta(x-y) \equiv \int \frac{d^d k}{(2\pi)^d} e^{ik(x-y)} \tilde{\Delta}(k^2)$$

• Free field theory

$$\tilde{\Delta}(k^2) = \frac{1}{k^2 + m^2 - i\epsilon}$$

• has isolated poles at $k^2 = -m^2$ w residue one.

m : physical mass of the particle

* Take $x^0 > y^0$ and insert a complete set of eigenstates

(a) Ground state $|0\rangle$ / Vacuum

(b) One-particle state $|k\rangle$, $\omega = (\vec{k}^2 + m^2)^{1/2}$

(c) Multiparticle continuum $|k, n\rangle$

$$\omega = (\vec{k}^2 + M^2)^{1/2}, \quad M \geq 2m$$

$$\begin{aligned} \langle 0 | \phi(x) \phi(y) | 0 \rangle &= \langle 0 | \phi(x) | 0 \rangle \langle 0 | \phi(y) | 0 \rangle \\ &+ \int \hat{d}k \langle 0 | \phi(x) | k \rangle \langle k | \phi(y) | 0 \rangle \\ &+ \sum_n \int \hat{d}k \langle 0 | \phi(x) | k, n \rangle \langle k, n | \phi(y) | 0 \rangle \end{aligned}$$

$$\langle k | \phi(y) | 0 \rangle = e^{-iky} \quad \langle 0 | \phi(x) | k \rangle = e^{ikx}$$

$$(\text{Sec. term}) = \int d\tilde{k} e^{-ik(y-x)} \\ = \int \frac{d^{d-1}k}{(2\pi)^{d-1}} 2\omega e^{-ik(y-x)}$$

* Since $\phi(x) = e^{-ip_M x_M} \phi(0) e^{ip_M x_M}$

$$\langle K, n | e^{ip_M x_M} \phi(0) e^{ip_M x_M} | 0 \rangle = e^{ip_M x_M} \langle K, n | \phi(0) | 0 \rangle$$

$$K^0 = (K^2 + M^2)^{1/2}$$

$\Leftrightarrow$

$$\langle \Omega | T(\phi(x) \phi(y)) | \Omega \rangle = \int d\tilde{k} e^{ik(x-y)} + \sum_n \int d\tilde{k} e^{ik(x-y)} |\langle K, n | \phi(0) | 0 \rangle|^2$$

Define the spectral density

$$\rho(s) \equiv \sum_n |\langle K, n | \phi(0) | 0 \rangle|^2 \delta(s - M^2)$$

$$\rho(s) \geq 0 \text{ for } s \geq 4m^2$$

$$\rho(s) = 0 \text{ for } s < 4m^2$$

$$\langle \Omega | \phi_x \phi_y | \Omega \rangle = \int d\tilde{k} e^{ik(x-y)} + \int_{4m^2}^{\infty} ds \rho(s) \int d\tilde{k} e^{ik(x-y)}$$

$\uparrow$
 $K^0 = (\vec{k}^2 + M^2)^{1/2}$

$\uparrow$
 $K^0 = (k^2 + s)^{1/2}$

* Now for $y^0 > x^0$

$$\langle \Omega | \phi_y \phi_x | \Omega \rangle = \int d\tilde{k} e^{-ik(x-y)} + \int_{4m^2}^{\infty} ds \rho(s) \int d\tilde{k} e^{-ik(x-y)}$$

• Using $\int \frac{d^d k}{(2\pi)^d} \frac{e^{ik(x-y)}}{k^2 + m^2 - i\epsilon} = \frac{i\theta(x^0 - y^0)}{2} \int d\tilde{k} e^{ik(x-y)} + \frac{i\theta(y^0 - x^0)}{2} \int d\tilde{k} e^{-ik(x-y)}$

$$\Rightarrow i \langle \Omega | T \phi_x \phi_y | \Omega \rangle =$$

$$\int \frac{d^d k}{(2\pi)^d} e^{ik(x-y)} \int \frac{1}{k^2 + m^2 - i\epsilon} + \int_{4m^2}^{\infty} ds \rho(s) \int \frac{1}{k^2 + s - i\epsilon}$$

$$\Delta(k^2)_{\pm} = \frac{1}{k^2 + m^2 - i\epsilon} + \int_{4m^2}^{\infty} ds \rho(s) \frac{1}{k^2 + s - i\epsilon}$$

(Lehmann - Källén form)

13.1 | $-\frac{1}{2} \int d^4x \partial^\mu \phi \partial_\mu \phi - \frac{1}{2} \int d^4x m^2 \phi^2 - \mathcal{L}_I$
 * Use the hint
 $[\phi(x), \dot{\phi}(y)]$

$$i \langle \Omega | \phi_x \dot{\phi}_y | \Omega \rangle = \dots$$

$$\langle \Omega | \phi_x \dot{\phi}_y | \Omega \rangle = \int \frac{d^d k}{(2\pi)^d} + i k^0 \left(\dots \right)$$

$$\langle \Omega | \phi_x \dot{\phi}_y | \Omega \rangle = i \int \frac{d^d k}{(2\pi)^3} 2 e^{i k(x-y)}$$

$$+ i \int_{4m^2}^{\infty} ds \rho(s) \int \frac{d^d k}{(2\pi)^3} 2 e^{i k(x-y)}$$

$$\langle \Omega | \phi_x \dot{\phi}_y | \Omega \rangle = \frac{i}{(2\pi)^3} \int d^d k e^{i k(x-y)} \left(1 + \int_{4m^2}^{\infty} ds \rho(s) \right)$$

$$= \frac{i}{2} \delta^d(x-y) \left(1 + \int_{4m^2}^{\infty} ds \rho(s) \right)$$

$$\langle \Omega | [\phi(x), \dot{\phi}(y)] | \Omega \rangle \Big|_{x^0=y^0}$$

$$= i \delta^d(x-y) \left(1 + \int_{4m^2}^{\infty} ds \rho(s) \right)$$

* Now, try to find the conjugate momenta of ϕ :
 i.e. $\Pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}$

$$\mathcal{L} = -\frac{1}{2} Z_\varphi \partial^\mu \varphi \partial_\mu \varphi - \frac{1}{2} Z_\varphi \partial^\mu \varphi \partial_\mu \varphi - \frac{1}{2} Z_m$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\varphi}} = Z_\varphi \partial_0 \varphi = Z_\varphi \dot{\varphi}$$

$$\Pi(y) = Z_\varphi \dot{\varphi}(y)$$

$$\langle \Omega | [\varphi(x), Z_\varphi^{-1} \Pi(y)] | \Omega \rangle$$

$$= Z_\varphi^{-1} \langle \Omega | [\varphi(x), \Pi(y)] | \Omega \rangle$$

$$= Z_\varphi^{-1} \langle \Omega | \Omega \rangle i \delta^d(\vec{x} - \vec{y})$$



$$Z_\varphi^{-1} i \delta^d(x - y) = i \delta^d(x - y) \left(1 + \int_{y_{\min}^2}^{\infty} ds \rho(s) \right)$$

$$\therefore Z_\varphi = \frac{1}{1 + \int_{y_{\min}^2}^{\infty} ds \rho(s)}$$